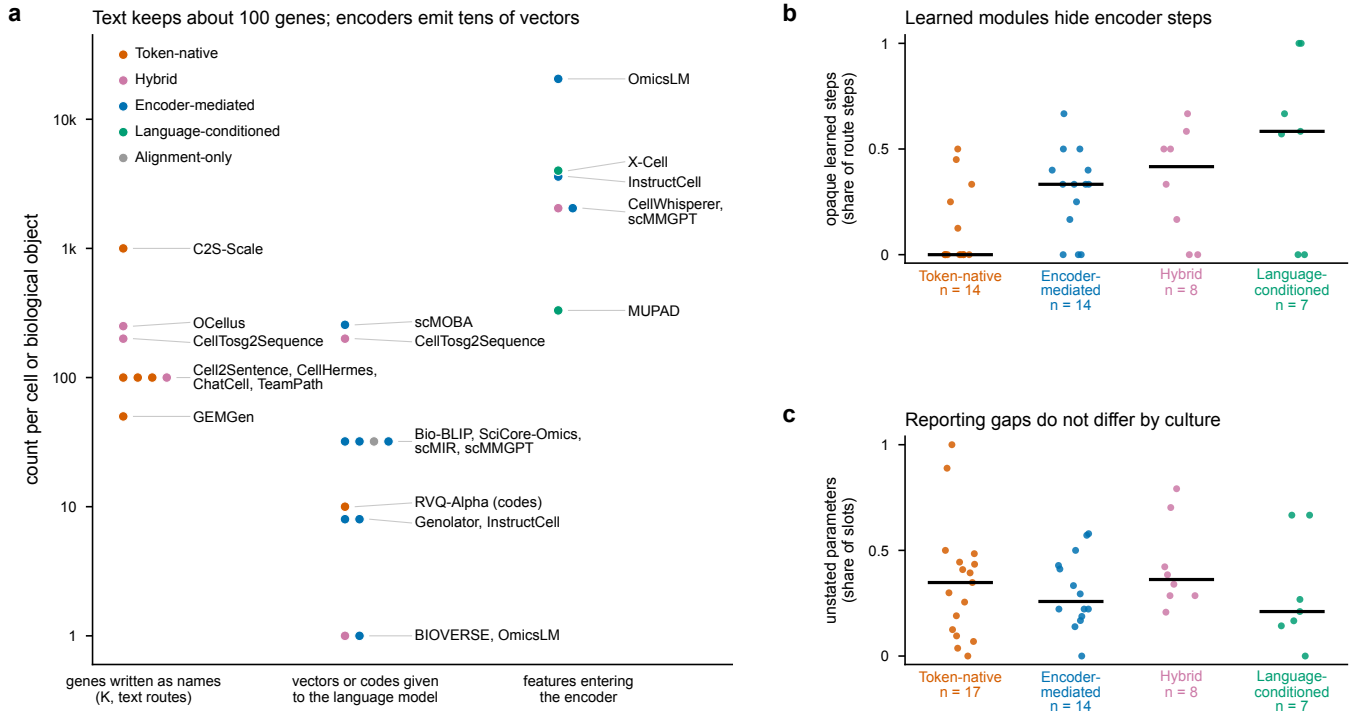
a
Text keeps about 100 genes; encoders emit tens of vectors
Token-native
Hybrid
Encoder-mediated
Language-conditioned
Alignment-only
count per cell or biological object
10k
1k
100
10
1
C2S-Scale
OCellus
CellTosg2Sequence
Cell2Sentence, CellHermes, ChatCell, TeamPath
GEMGen
scMOBA
CellTosg2Sequence
Bio-BLIP, SciCore-Omics, scMIR, scMMGPT
RVQ-Alpha (codes)
Genolator, InstructCell
BIOVERSE, OmicsLM
OmicsLM
X-Cell
InstructCell
CellWhisperer, scMMGPT
MUPAD
genes written as names (K, text routes)
vectors or codes given to the language model
features entering the encoder
b
Learned modules hide encoder steps
opaque learned steps (share of route steps)
1
0.5
0
Token-native n = 14
Encoder-mediated n = 14
Hybrid n = 8
Language-conditioned n = 7
c
Reporting gaps do not differ by culture
unstated parameters (share of slots)
1
0.5
0
Token-native n = 17
Encoder-mediated n = 14
Hybrid n = 8
Language-conditioned n = 7